

Experiment 11

VPN Configuration & Encryption – Report

Lab 11 Objectives:

- Configure a Virtual Private Network (VPN) and analyze its encryption mechanisms.

Prerequisites: The prerequisites of this lab are-

- Basic understanding of VPN concepts and encryption techniques.
- Familiarity with network security principles and tunneling protocols.

Outcomes: By the end of this lab, students will be able to:

- Configure and establish a VPN connection.
- Analyze encryption mechanisms used in VPN tunnels.
- Evaluate security, performance, and data integrity in a VPN setup.

Introduction

A **Virtual Private Network (VPN)** allows secure communication between two or more devices over an insecure network like the internet. It creates an encrypted tunnel that protects data from interception, ensuring confidentiality and integrity. VPNs use encryption algorithms and tunneling protocols to secure data.

Experiment Description

1. Setting up VPN Server and Client:

The VPN server and client were configured using software like OpenVPN, IPSec, or WireGuard. Necessary authentication keys and certificates were generated for secure communication.

2. Establishing Secure VPN Tunnel:

After configuration, a VPN tunnel was established between the two

devices. This tunnel encrypted all traffic, preventing unauthorized access or eavesdropping.

3. **Capturing VPN Traffic:**

Network packets were captured using **Wireshark** or **tcpdump** during VPN communication. The captured packets showed encrypted payloads, proving data confidentiality.

4. **Testing Data Transmission Security:**

Tests like ping and file transfer were conducted over the VPN tunnel to ensure data integrity and security during transmission.

5. **Performance Comparison:**

Network speed and latency were tested with VPN encryption enabled and disabled to evaluate the impact of encryption on performance.

6. **Documentation:**

All configuration details, packet capture analyses, and performance observations were documented.

Encryption Mechanisms in VPN

- **AES (Advanced Encryption Standard):** A symmetric key encryption algorithm used to encrypt VPN traffic. AES-256 provides strong encryption security.
- **HMAC (Hash-based Message Authentication Code):** Ensures integrity and authenticity of packets, protecting against tampering.
- **TLS (Transport Layer Security):** Provides secure key exchange and authentication between VPN server and client.
- **RSA:** Often used for secure authentication and key exchange in VPN protocols.

Tools Used

- **VPN Software:** OpenVPN, WireGuard, or IPSec-based VPNs for setting up VPN connections.
- **Packet Capture Tools:** Wireshark or tcpdump for capturing and analyzing VPN traffic.
- **Command-Line Utilities:** openvpn, ipsec, netstat for configuration and monitoring.

Observations and Analysis

- The VPN tunnel encrypted data packets, making payloads unreadable in packet captures.
- AES-256 encryption and HMAC-SHA256 were used to secure and verify data.
- TLS handshake ensured safe exchange of keys and client/server authentication.
- Performance tests showed minor latency increase and bandwidth reduction due to encryption overhead.
- Data transmission through VPN was secure and reliable, confirmed by successful tests.

VPN Tunnel Diagram

